

5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

CODE:

Cl

```
c; clear; close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000; % High-pass cutoff frequency (Hz)
```

```
fs = 8000; % Sampling frequency (Hz)
```

```
N = 50; % Filter order
```

```
wc = fp/(fs/2); % Normalized cutoff (0–1)
```

```
%% --- FIR High Pass using Hamming Window ---
```

```
b_hamm = fir1(N, wc, 'high', hamming(N+1));
```

```
disp('--- High-Pass Coefficients (Hamming Window) ---');
```

```
disp(b_hamm);
```

```
%% --- Frequency Response Plots ---
```

```
figure;
```

```
freqz(b_hamm,1);
```

```
title('High-pass FIR using Hamming Window');
```

```
title('High-pass FIR using Hanning Window');
```

OUTPUT:

